

+ O₂ + electron (freeing O₂ at the cathode)

Depending on the type of electrolyte used, lithium-air batteries can be grouped into four categories (see Fig. 4):

- (a) Aprotic (using non-aqueous electrolyte)
- (b) Aqueous
- (c) Hybrid: (a) + (b) or (a) + (d)
- (d) Solid-state

Aprotic electrolyte (non-aqueous) battery

As shown in Fig. 5, a typical aprotic lithium-air battery contains a metal lithium anode, an electrolyte comprising a dissolved lithium salt in an aprotic solvent, a conducting porous carbon supported oxygen-breathing cathode, and often a lithium-ion conducting membrane separating the cathode and anode. The destabilisation of the lithium metal interface by crossover of water, oxygen, and carbon-dioxide can fatally degrade the cycle life of lithium-air battery. Hence, usually, polyurethane separators are used that can effectively suppress this crossover while allowing lithium ions to diffuse through selectively. The membrane separator is a key component in a liquid lithium-air battery for electrically separating the cathode and the anode, meanwhile ensuring ionic transport between them and also protect the lithium metal anode from redox mediators.

The major discharge product at the cathode is lithium-peroxide (Li₂O₂) with a small proportion of lithium-oxide (Li₂O). Reaction product Li₂O is not desirable, because it is not reversible, being

not fully rechargeable back to lithium and oxygen. Therefore, Li₂O₂ is considered the ideal discharge product at the cathode.

Here lithium metal is the typical anode choice, which however risks dendritic lithium deposits triggering a short circuit. During operation, layers of lithium salts are deposited onto the anode. This barrier eventually inhibits the reaction kinetics between the anode and the electrolyte. Earlier the commonly used electrolyte was lithium salt dissolved in ethylene-carbonate plus propylene-carbonate solvents but, unfortunately, they are also inflammable. However, such solvents are not stable to the superoxide radicals and could be decomposed during charge and discharge process, leading to fast termination of the battery.

Currently, the most common sol-

vents used are ethers and sulphonate based solvents, which have long-term stability. Lithium salts like LiBF₄, LiBOB (lithium bis borate), and LiTFSI (a hydrophilic salt), which have excellent stability and high ionic conductivity, are dissolved in these solvents to make the electrolyte. Lithium-air batteries with 610 cycles in ambient air have been made by combination of low-density polythelene film that prevents water erosion and gel electrolyte that contains redox mediator.

Gel polymer electrolyte (GPE) consists of polymer matrix like polythelene oxide or its derivatives, lithium salts, and plasticisers. It combines the advantages of both solid and liquid components. GPEs possess outstanding ionic conductivity, which can boost the electrochemical performance and have good interfacial

properties from the liquid phase as well as good mechanical properties from solid component.

As the liquid electrolyte is immobilised in the pores of polythene matrix the risk of leakage of the liquid is reduced manifold compared to commercial separators. It prevents the detrimental side reaction of discharged Li₂O₂ to Li₂CO₃ in ambient air, thus facilitating electrochemical decomposition of Li₂O₂ during charging, which improves the reversibility of the battery.

Another method uses lithium anode with a passivation layer of Li₂CO₃/C coating by treating with pure carbon-dioxide, which prevents the degradations of lithium metal from side reactions. This method achieves excellent cycling performance over 700 cycles, equivalent to several years of use. A

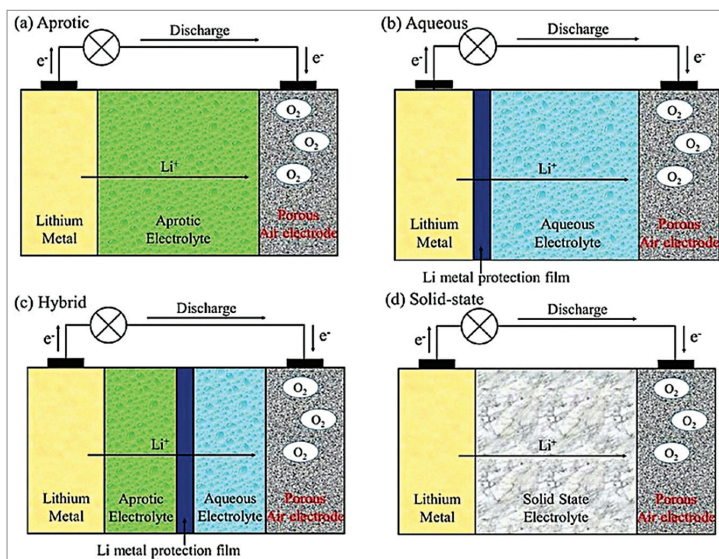


Fig. 4: Variants of lithium-air batteries

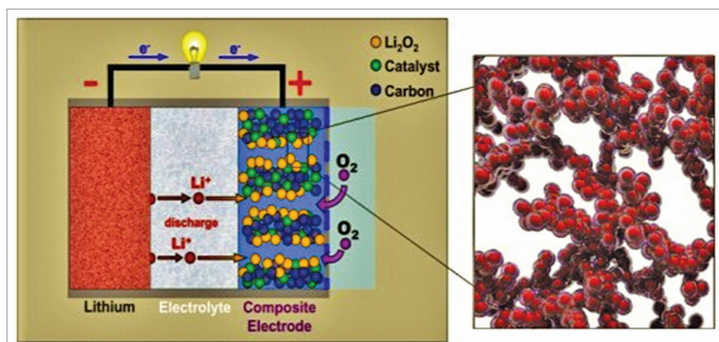


Fig. 5: Aprotic electrolyte lithium-air battery